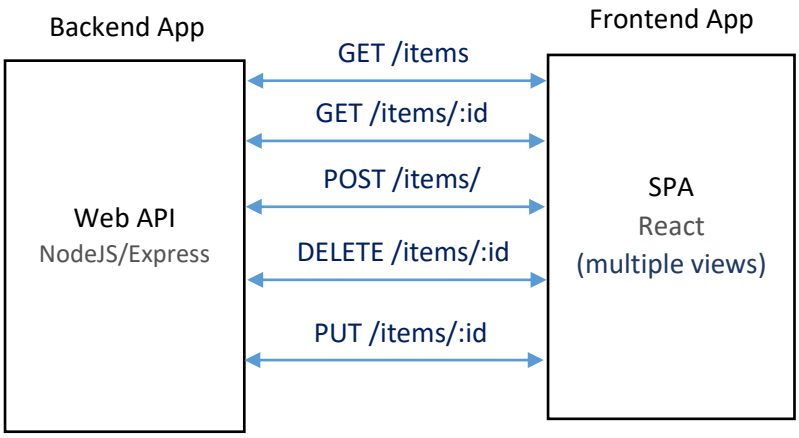
CSC457 – Mobile App Development
Project 1

Course/Student Learning Outcomes: 1, 4, 5

The objective of this project is to create a Web application consisting of a Backend app and Frontend app as depicted in the below diagram.



1- Backend Web App: (20 points)

The backend app should consist of a Web Server developed in NodeJS/Express to expose a Web API for managing one entity type of your choice (fruits, products, books, orders, ...etc.). The Web API should provide routes to perform CRUD operations over an in-memory dataset of your chosen entity type. The CRUD operations are performed using HTTP Get (for reading), HTTP Post (for creating), HTTP Delete (for deleting), and HTTP Put (for updating). The chosen entity type should include at least 5 properties of different data types.

Note: This backend app should not involve any database logic, authentication/authorization logic, nor business logic. It should be a bare bone Web API for performing CRUD operation over an in-memory dataset of a particular entity type.

2- Frontend Web App: (80 points)

The frontend app should consist of a *Single Page Application (SPA)* developed in React which provides a UI for connecting to the previously developed Backend app and managing the chosen entity type. It should include the following views:

- *Home view*,
- *About us view*,
- *Contact us view,*
- *Display all entities view*.

- *Display a single entity view,*
- *Submit a new entity view,*
- *Delete an existing entity view, and*
- *Update an existing entity view.*

Details:

- You should make effective use of React Components, Props, and States.
- Look and feel, and design (UI/UX) is very important.
- *Contact us* view should include a Form with Event Handling and Input Validation.
- The different entity views (display all, display one, submit, delete, update) should make use of the Web API you developed previously by sending the necessary HTTP requests in order to perform the CRUD operations over the chosen entity type.

Required Research Steps:

- You should research and use ***"React Router"*** library to implement client-side routing (navigation) between the different views (Home, About us, Contact us,…). Documentation and tutorials for the ***"React Router"*** can be found on the following link: https://reactrouter.com/en/main/start/tutorial

- You should research and use ***"Redux"*** library for managing and centralizing your application state. Further information about ***"Redux"*** can be found on the following link: https://redux.js.org/introduction/getting-started